



12B+13

LIQUIDS & SOLIDS+SOLUTIONS

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 13

1. Predict the solubility or miscibility of one substance in another.
2. Describe the energetics of the process of solution formation.

CHAPTER 12

1. List and describe the four types of intermolecular forces and identify the predominant intermolecular force in a substance.
2. Define and apply terms related to phase changes.
3. Calculate energy changes related to phase changes.
4. Interpret heating curves.
5. Predict physical properties (relative vapor pressures and relative phase change temperatures) based on strengths of intermolecular forces.
6. Define, describe, and interpret phase diagrams.
7. Describe the structures of various cubic unit cells.

Intermolecular Forces

12.1

*Types of IMFS
depend mostly on
molecular polarity!*

Generally, IMF strength increases

	LDF	DD	HB	ID
	London Dispersion Forces	Dipole-Dipole Attractions	Hydrogen Bonding Interactions	Ion-Dipole Attractions
Nonpolar molecules + noble gases	dominant	×	×	×
Polar molecules	✓	dominant	×	×
Polar molecules that have H—O, H—N, or H—F bonds	✓	✓	dominant	×
Polar molecules + Soluble ionic compounds	✓	?	?	dominant

The Solution Process

13.1

In *aqueous* solutions, *water* molecules (the solvent) surround solute particles in a process called hydration.

- We say that the ions are *hydrated*.
- A more general term, solvation, applies to any solvent.

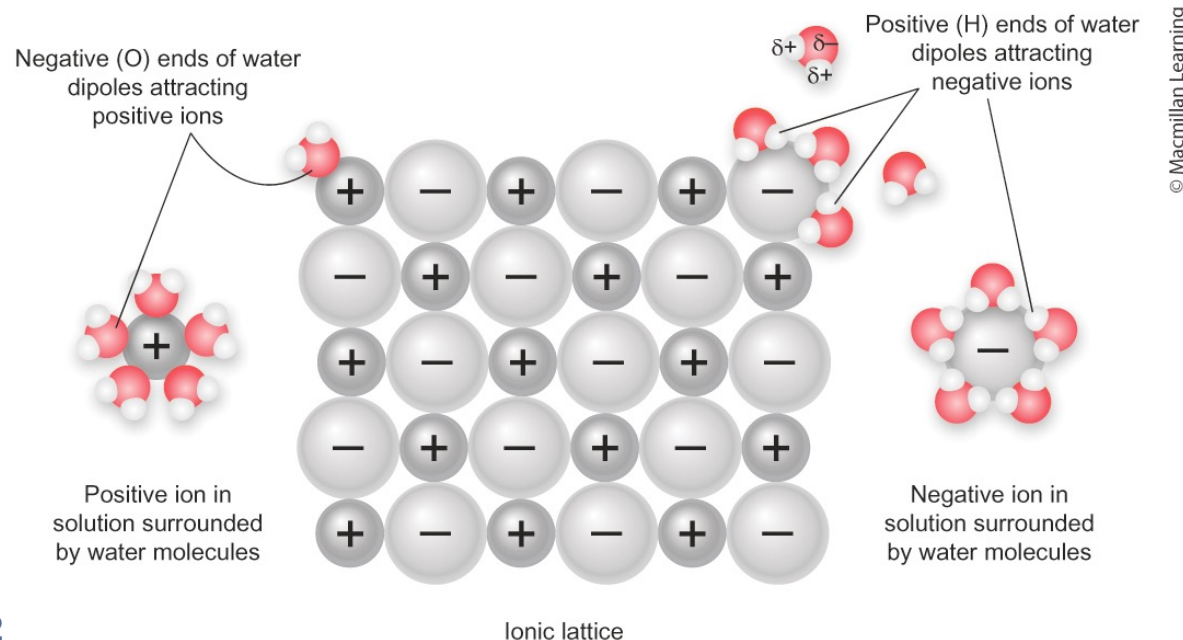


FIGURE 13.2

The Solution Process

13.1

A solution is a homogeneous *mixture* of almost any two phases of matter.

- The dissolved substance is the solute, often the minor component.
- The substance it is dissolved in is the solvent, often the major component.

Solution Type	Examples
Solid in solid	Alloys, like brass (Zn and Cu)
Solid in liquid	Aqueous solutions, like sugar water
Liquid in solid	Mercury in silver (dental amalgam)
Liquid in liquid	Gasoline (mixture of hydrocarbons)
Gas in solid	Hydrogen (H ₂) in platinum
Gas in liquid	Carbonated drinks (CO ₂ in water)
Gas in gas	Air (O ₂ in N ₂)

TABLE 13.1

The Solution Process

13.1

Substances **dissolve** in one another if the solute and solvent are able to form intermolecular attractions with each other.

- We often refer to the “*like dissolves like*” rule.
- In other words, substances with the *same* or *similar* types of IMFs tend to dissolve in each other.



polar solute in polar solvent ←

nonpolar solute in nonpolar solvent ←

IN-CLASS QUESTION 1

Learning Objective(s): 13.1

Select all of the compounds you would expect to be soluble in $\text{C}_6\text{H}_6(\ell)$.

Hint: Consider Lewis structures.

☐ H_2O

polar

☒ C_8H_{18}

nonpolar

☒ I_2

nonpolar

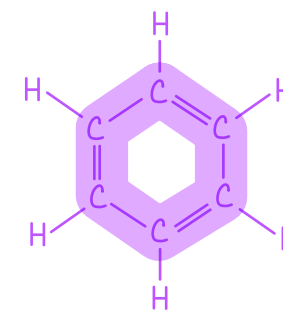
☐ Na_2SO_4

ionic $\leftarrow$ soluble in polar solvents via ion-dipole

match polarity and IMFs
of solute and solvent
so the pair can interact

nonpolar

LDF



HYDROCARBONS (C_xH_y) OFTEN NONPOLAR MOLECULES

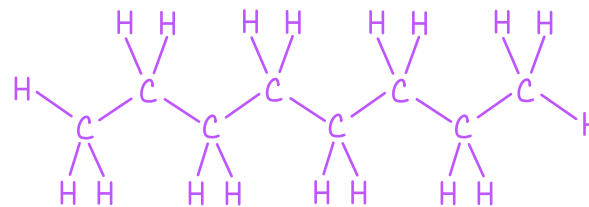
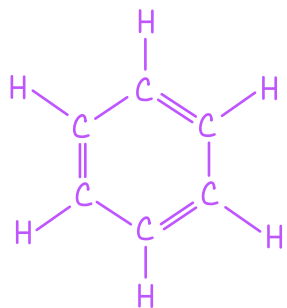
Chemistry in Context

There are two *necessary conditions* for a molecule to be polar:

1. The molecule must contain polar bonds so that bond dipoles exist.
2. The bond dipoles must not cancel out symmetrically due to the molecular shape.

Most hydrocarbon molecules (C_xH_y) are nonpolar molecules because they are comprised of *mostly nonpolar covalent bonds* (condition 1).

- The CC bonds are nonpolar covalent bonds because $\Delta EN = 0$
- The CH bonds are *approximately* nonpolar covalent bonds because $\Delta EN \approx 0$

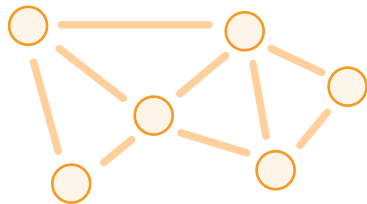


The Solution Process

13.1

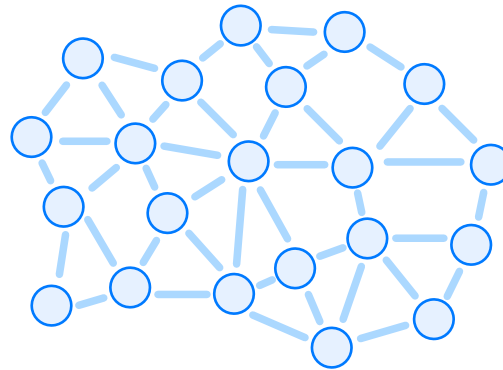
For a solution to form, consider three factors:

break
solute-solute attractions



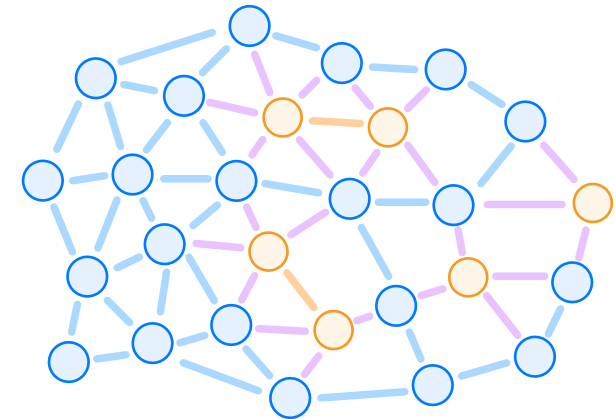
+

break
solvent-solvent attractions



→

form
solute-solvent attractions



For a solution to form: solute-solvent attractions must be greater than or equal to solute-solute and solvent-solvent attractions.

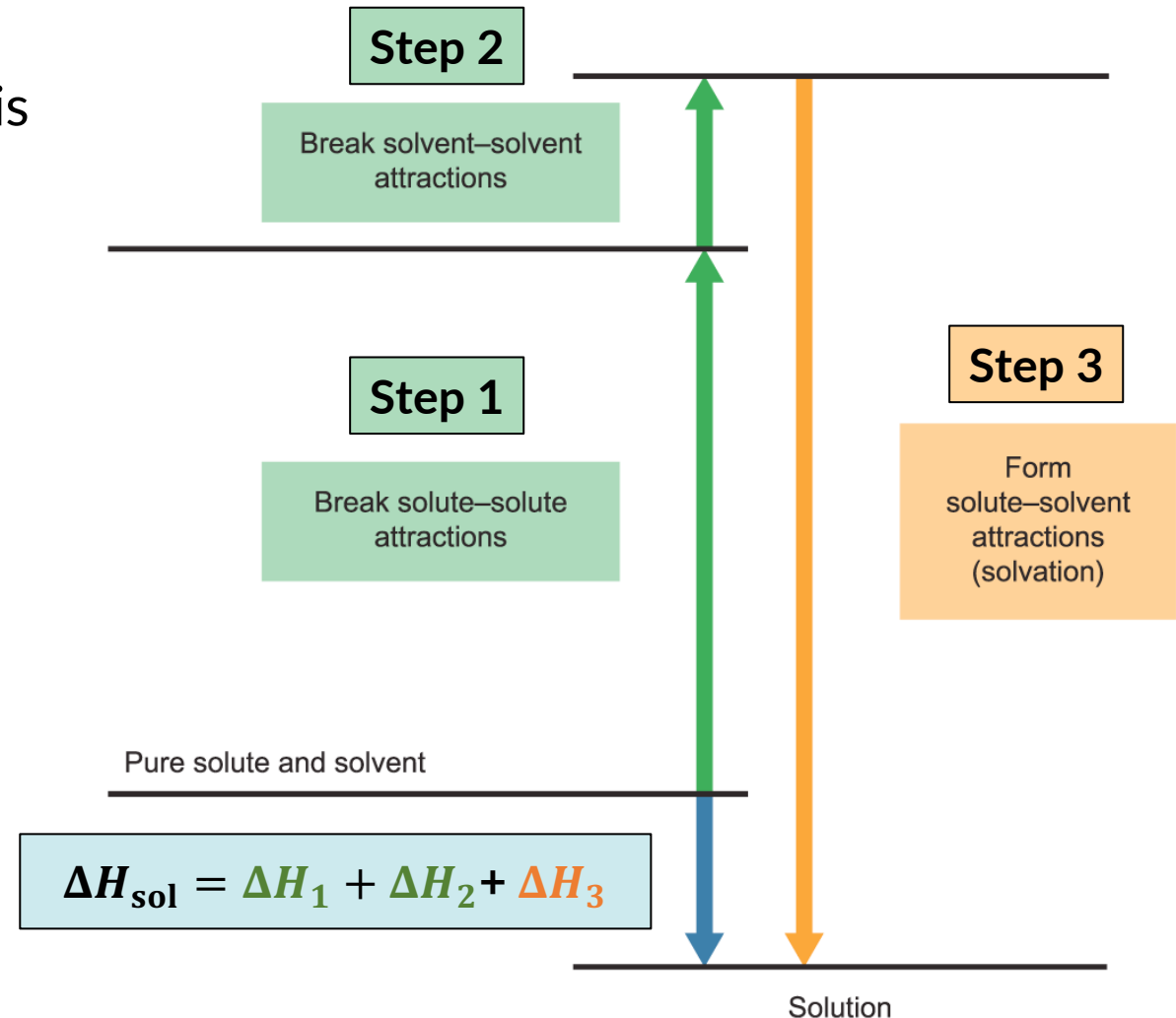
The Solution Process

13.1

The enthalpy of solution, ΔH_{sol} , is the overall energy change during solution formation and is the *sum* of the three steps (via Hess's Law).

Steps 1 and 2 are **endothermic**, requiring an input of energy proportional to the strength of the IMFs or lattice energy.

Step 3 is **exothermic**, releasing energy as the *new* solute-solvent attractions form.



The Solution Process

13.1

Solutions are *more likely to form* when $\Delta H_{\text{sol}} < 0$ (exothermic), but this is not a necessary conditions, so it is possible for $\Delta H_{\text{sol}} > 0$ (endothermic).

Entropy, the degree of *randomness* or *disorder* in a system, is also a factor in solution formation.

- Solutions are typically *more disordered* than pure solids and pure liquids.
- Formation of a solution always results in an *increase* in entropy of the system.

LEARNING OBJECTIVES

CHAPTER 12

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Phase Changes and Heating Curves

12.3

Each phase change has a corresponding change in enthalpy, ΔH , that can be used to determine the heat (q) associated with that phase change.

- *Breaking* attractive interactions (like IMFs) *requires energy (endothermic)*.

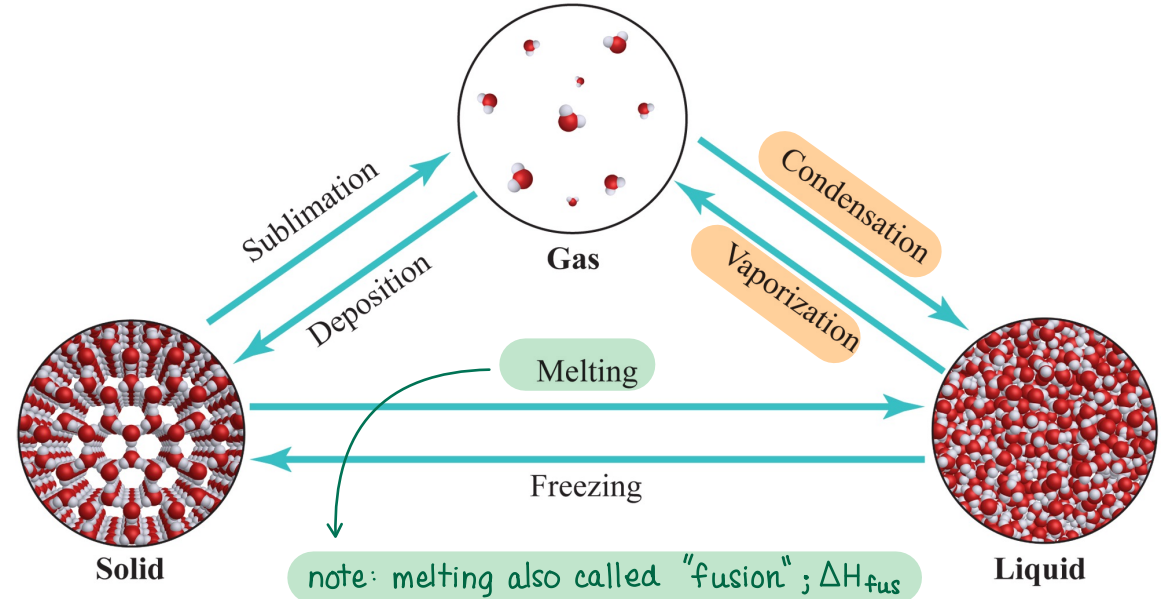
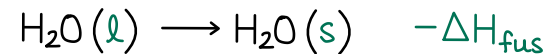
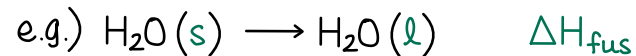
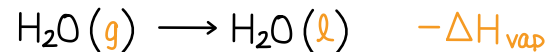
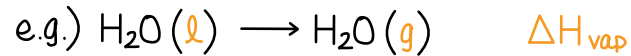


FIGURE 12.4

Phase Changes and Heating Curves

12.3

During a phase change there is no temperature change: $\Delta T = 0$

- To find heat (q) for the phase transition, the ΔH must be used.

$$q = n \cdot \Delta H$$

of mol

How much energy (in kJ) is needed to *evaporate* 50.0 g H_2O if $\Delta H_{\text{vap}} = 40.7 \text{ kJ/mol}$?



$$q = n \cdot \Delta H$$

$$= 2.775 \text{ mol} \cdot 40.7 \frac{\text{kJ}}{\text{mol}}$$

$$q = 113 \text{ kJ}$$

$$n = 50.0 \text{ g H}_2\text{O} \times \frac{1 \text{ mol H}_2\text{O}}{18.015 \text{ g H}_2\text{O}} = 2.775 \text{ mol H}_2\text{O}$$

Phase Changes and Heating Curves

12.3

Heating curves are graphs showing the variation in the temperature of a sample as it is heated at constant rate and constant pressure.

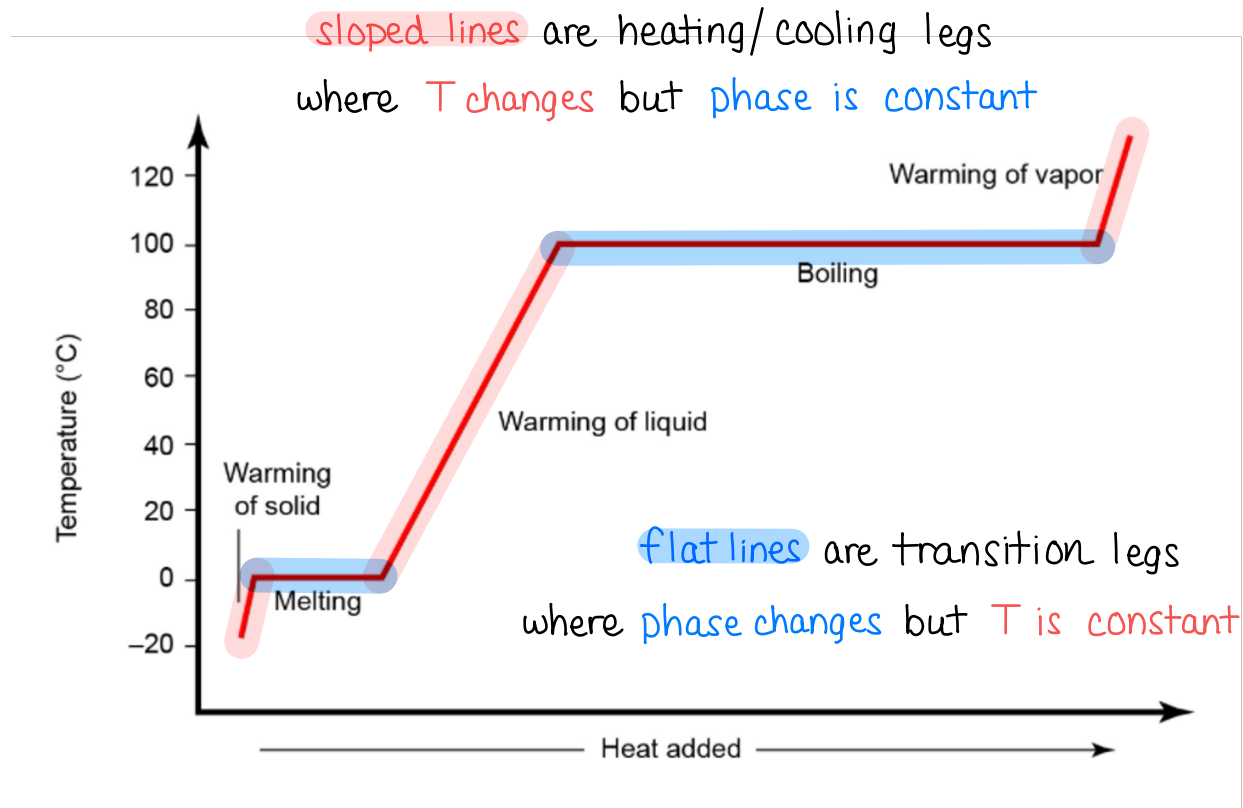


FIGURE 12.16

heating/cooling legs:

$$q = m c_{\text{solid}} \cdot \Delta T$$

$$q = m c_{\text{liquid}} \cdot \Delta T$$

$$q = m c_{\text{gas}} \cdot \Delta T$$

notes: ΔT gives sign of q , c depends on phase

transition legs:

$$q = n \cdot \Delta H_{\text{vap}}$$

$$q = n \cdot \Delta H_{\text{fus}}$$

notes: need to flip sign of ΔH_{vap} or ΔH_{fus} if reverse phase transitions (gas $\rightarrow$ liquid $\rightarrow$ solid)

IN-CLASS QUESTION 2

Learning Objective(s): 12.3

Calculate the total heat (in units of kJ) transferred in the conversion of 20.0 g of water vapor (steam) at 104.0 °C to water at 10.5 °C.

$$n = 20.0 \text{ g H}_2\text{O} \times \frac{1 \text{ mol H}_2\text{O}}{18.015 \text{ g H}_2\text{O}}$$

$$n = 1.11 \text{ mol H}_2\text{O}$$

$$c_{\text{H}_2\text{O}(s)} = 2.087 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}$$

$$c_{\text{H}_2\text{O}(l)} = 4.184 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}$$

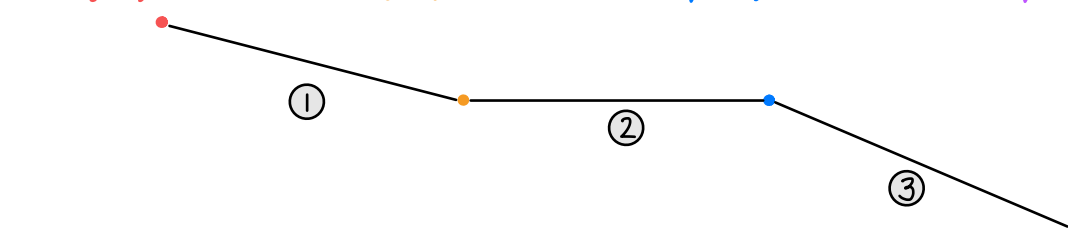
$$c_{\text{H}_2\text{O}(g)} = 2.042 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}}$$

$$\Delta H_{\text{fus}} = 6.01 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta H_{\text{vap}} = 40.7 \frac{\text{kJ}}{\text{mol}}$$

notes: units mismatch,
c has J and g,
ΔH has kJ and mol,
flip sign on ΔH b/c cooling

gas, 104.0 °C → gas, 100.0 °C → liquid, 100.0 °C → liquid, 10.5 °C



① gas, 104.0 °C → gas, 100.0 °C

$$q_1 = m \cdot c_g \cdot \Delta T = (20.0 \text{ g}) \left(2.042 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}} \right) (100.0^\circ\text{C} - 104.0^\circ\text{C}) \times \frac{1 \text{ kJ}}{1000 \text{ J}} = -0.163 \text{ kJ}$$

② gas, 100.0 °C → liquid, 100.0 °C

$$q_2 = n \cdot (-\Delta H_{\text{vap}}) = (1.11 \text{ mol}) \left(-40.7 \frac{\text{kJ}}{\text{mol}} \right) = -45.18 \text{ kJ}$$

③ liquid, 100.0 °C → liquid, 10.5 °C

$$q_3 = m \cdot c_l \cdot \Delta T = (20.0 \text{ g}) \left(4.184 \frac{\text{J}}{\text{g} \cdot ^\circ\text{C}} \right) (10.5^\circ\text{C} - 100.0^\circ\text{C}) \times \frac{1 \text{ kJ}}{1000 \text{ J}} = -7.489 \text{ kJ}$$

gas, 104.0 °C → liquid, 10.5 °C

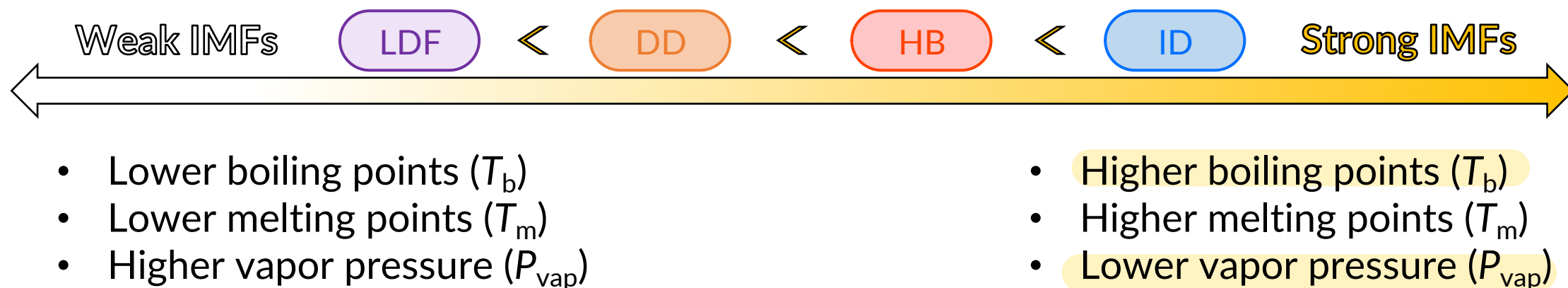
$$q_{\text{tot}} = q_1 + q_2 + q_3 = -52.8 \text{ kJ}$$

Intermolecular Forces

12.1

Physical molecular properties can be predicted from the types and strengths of IMFs.

Only IMFs are broken during phase changes (no chemical bonds are broken).

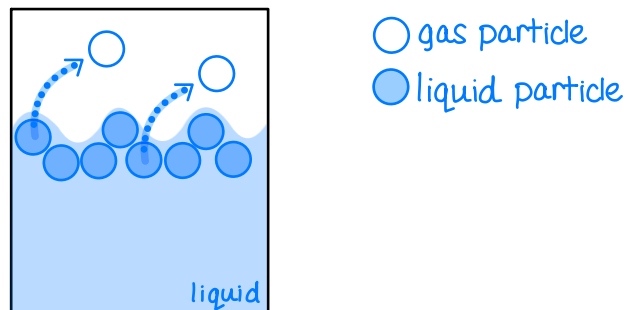


Vapor Pressure and Boiling Point

12.4

Vapor pressure is the pressure exerted by a gas on the surface of the liquid from which it just evaporated.

- IMFs in the liquid phase must be overcome for evaporation to occur.



- Volatile* substances vaporize easily due to weaker IMFs
- Nonvolatile* substances do not easily vaporize due to stronger IMFs

Vapor Pressure and Boiling Point

12.4

Boiling point

- The temperature at which the vapor pressure for a substance is equal to the atmospheric pressure.

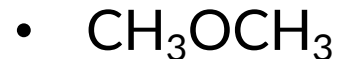
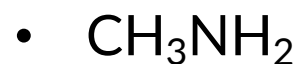
Normal boiling point

- The temperature at which the vapor pressure for a substance is 1.00 atm.

EXAMPLE PROBLEM

Learning Objective(s): 12.5

Arrange the following pure samples in order of *decreasing vapor pressures* and *increasing boiling points*.

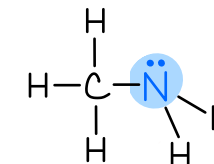
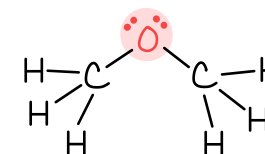
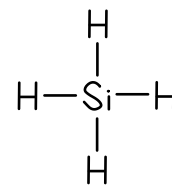
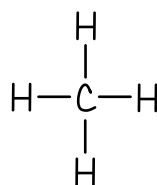
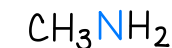


weak IMFs
low boiling point
high vapor pressure

$$P_{\text{vap}}: \text{Ne} > \text{CH}_4 > \text{SiH}_4 > \text{CH}_3\text{OCH}_3 > \text{CH}_3\text{NH}_2$$

$$T_b: \text{Ne} < \text{CH}_4 < \text{SiH}_4 < \text{CH}_3\text{OCH}_3 < \text{CH}_3\text{NH}_2$$

strong IMFs
high boiling point
low vapor pressure



n/a

nonpolar

nonpolar

polar

polar

LDF

LDF

LDF

DD

HB

same dominant IMF;
compare polarizability: size, mass, e⁻ count

BONUS QUESTION

Assessed on: Fall 2021, Exam 3

Which solutes do you expect to be soluble in H_2O ?

Select all that apply.

